

Khang Tran

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Education

Princeton University, BA in Physics; minors in Computer Science and Engineering – Princeton, NJ Sept 2024 – May 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Machine Learning & Pattern Recognition, Neural Rendering, Computer Graphics, Algorithms & Data Structures, Honors Linear Algebra

NUS High School of Mathematics and Science, Diploma (High Distinction) in Engineering & Computer Science – Singapore Jan 2020 – Oct 2023

- **GPA:** 4.9/5.0 — Computer Organization, Operating Systems (NUS dual enrollment)

Experience

Undergraduate Researcher, Roberto Car Group, Princeton University – Princeton, NJ June 2026 – present

- Built a fault-tolerant, auto-relaunching pipeline that runs large-scale hybrid-DFT path-integral MD (192 atoms, 8 ring-polymer beads) across multi-node HPC (SLURM) clusters, automatically recovering from job timeouts and failures, to generate the reference dataset used to train the group's machine-learned interatomic potentials (DeepMD).
- Automated job orchestration and parallelization across nodes to scale training-data throughput.

Research Intern, Climate Modeling, High Meadows Environmental Institute, Princeton University – Princeton, NJ June 2025 – Aug 2025

- Analyzed large climate-model ensemble datasets in Python, quantifying how varying CO2 forcing strengths affect Arctic Amplification under CO2 removal.

Space Systems Engineer Intern, Aliena – Singapore Oct 2023 – Mar 2024

- Built an end-to-end Python automation pipeline for remote control of laboratory instrumentation during ignition testing of 5A hollow cathodes, replacing manual operation.
- Designed and built a hardware-in-the-loop board emulating a Hall thruster to test its ECU software.

Projects

2D-to-3D Style Transfer for Reusable Assets Feb 2026 – May 2026

COS526: Neural Rendering (Princeton, Prof. Felix Heide)

- Developed a diffusion-based pipeline that stylizes a 3D mesh from a single reference image, baking PBR material maps (albedo, normal, roughness) via differentiable rendering.
- Built the multiview-consistent style-transfer stage in PyTorch — SDXL + ConsisLoRA (LoRA fine-tuning) with depth/Canny ControlNets, plus a RePaint inpainting schedule and BiRefNet masking to enforce cross-view consistency.

Failure-Mode Analysis of ML Interatomic Potentials on an Ab-Initio Reference Surface May 2026

Independent study; poster, Princeton AI Fluency Lab (May 2026)

- Compared the H₃ potential-energy surface ($H + H_2 \rightarrow H_2 + H$) from **six** pretrained ML potentials (MACE-MP-0, CHGNet, M3GNet, UMA, ORB, SevenNet), run as ASE calculators, against the BKMP2 *ab initio* reference (**5.1–72.4 kcal/mol** MAE) — all six predicted an **incorrect transition state**, and only two recovered the minimum-energy path even when given the true one.

Skills

Languages: Python, C/C++, MATLAB, Mathematica, LaTeX

ML & Data: PyTorch, NumPy/SciPy, distributed/parallel computing, HPC pipelines (SLURM), data-generation automation, DFT

Areas: Deep learning, diffusion/generative models, neural rendering, scientific/numerical computing

Teaching & Leadership: CS Lab TA, COS217/226 (2026) & Physics Tutor, Princeton Physics Dept. (2025–26); Problem Setter & Organizer, Princeton Science Olympiad (2025–26); Problem Setter, Singapore Physics League / SPhL (2023–25)

Awards: Pyka Prize for Physics Excellence, Princeton (2026); IBM Qiskit Hackathon Champion (Hardware track); COSCON 2025 1st (Undergrad), COSCON 2024 5th; Singapore Physics Olympiad Gold (2022)